

Mechanical Properties of Materials

Recall stress as the amount of force per unit area, or

$$\sigma = \frac{F}{A}$$

More specifically, this is the normal stress experienced by a material, and can either be tensile or compressive

When an object deforms, it undergoes strain, as

$$\varepsilon = \frac{\Delta L}{L}$$

expressed as a dimensionless quantity.

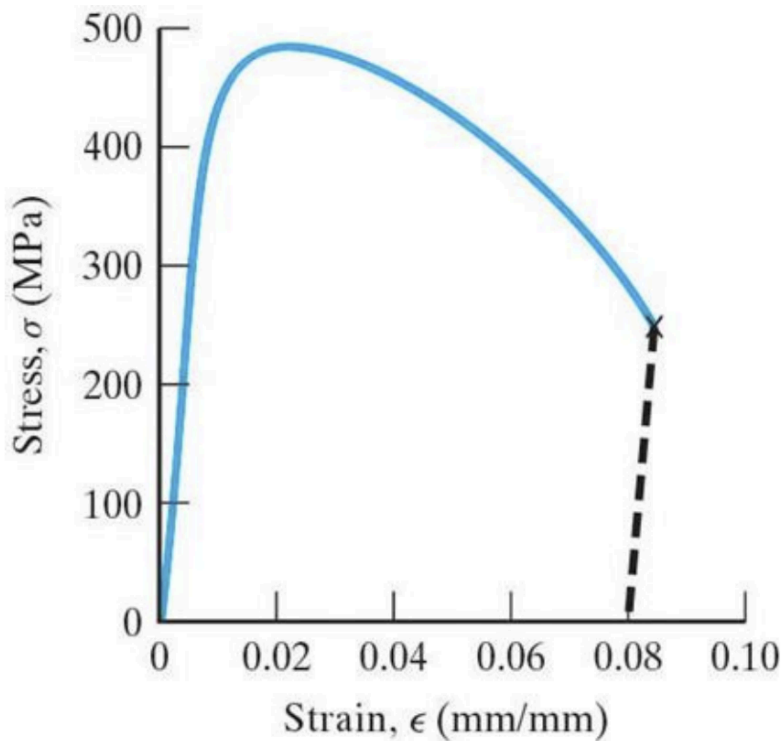
Under small amounts of strain, the stress experienced is directly proportional to the strain, as the material is undergoing elastic deformation, and its deformation can be explained by Hooke's Law.

Hence,

$$\begin{aligned}\sigma &\propto \varepsilon \\ \sigma &= E\varepsilon\end{aligned}$$

where the proportionality constant is Young's Modulus.

Stress, strain and Young's Modulus can be found via a tensile test, which outputs the force per extension, to be normalised to form a stress-strain curve.

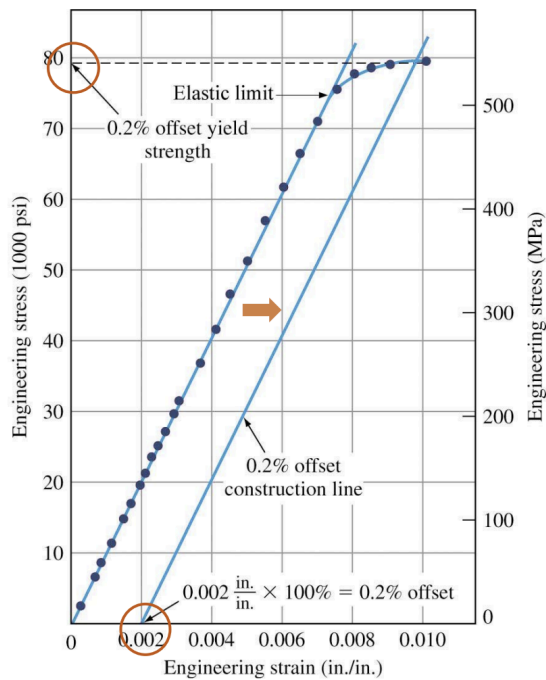


On the topic of the Young's Modulus, the start of the stress-strain curve is linear, where the slope is E .

Young's Modulus is also the stiffness of the material; how much it resists elastic deformation. For mechanical systems, adequate stiffness is required, especially for things that move.

After elastic deformation, the material undergoes plastic deformation, where the deformation is permanent. The process of switching between the two modes of deformation is called yielding. In some materials, there is usually an obvious drop in the stress, known as the yield point.

However, for materials that do not have an obvious yield point, we use 0.2% or 0.002 strain as a reference to find the yield strength.

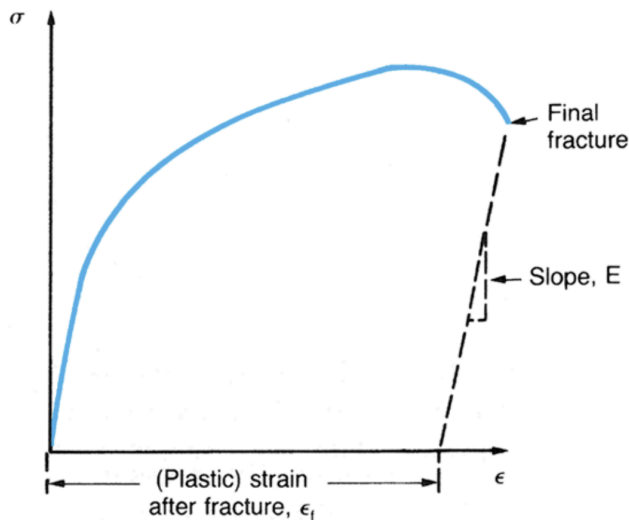


By drawing a line parallel to the elastic deformation portion of the curve, that passes through 0.002 strain, the intersection with the curve is the 0.2% offset yield strength, written as σ_y .

Knowing the yield strength, we can ensure that our loading conditions do not surpass the yield strength. Typically, we employ a factor of safety to prevent unexpected loading from deforming the material plastically.

As the material continues to be pulled, it reaches its ultimate tensile stress, UTS, represented as the maximum point on the stress-strain curve. UTS is usually used as a metric of material quality, as internal pores or inclusions reduce UTS.

Other than strength, we can also determine other material properties from the stress-strain curve.



Ductility can be determined from the stress-strain curve as the amount of plastic deformation, or the percentage elongation, $\%EL$.

Brittle materials will have the opposite, having a lesser span and lesser percentage elongation. We can quantify brittle materials as having less than 5% of strain (0.005 strain) at fracture.

Recall the DBTT from materials 2, where as the temperature decreases, brittleness increases (or ductility decreases), however strength tends to increase.

A similar quantity is the toughness of a material, referring to the amount of energy a material can absorb under impact, quantified as joules.

As the area under the curve is the work done to deform the material, it also represents the energy needed to fracture the material, giving the toughness of the material. Because of this, tougher materials typically have higher strength and ductility. Tougher materials also have more deformation at the fracture zone.

Other metrics cannot be fully determined by the tensile test. One of them is hardness.

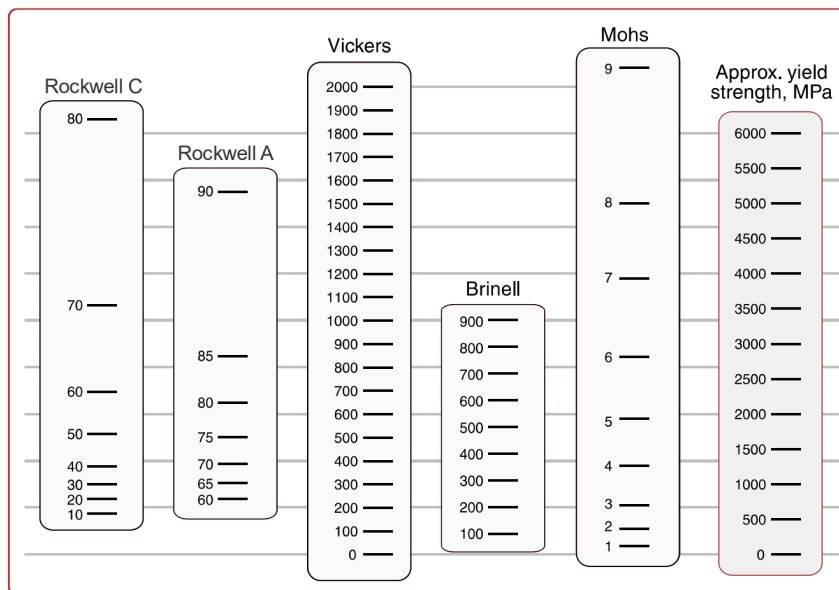
Hardness is a material's resistance to surface indentation. Hence, a hardness test involves using an indenter to press into the material, and measuring the effects, like shape or depth of the indentation.

As the hardness test involves plastically deforming the material, there is a positive relationship between hardness and yield strength. Thus, a hardness

test may be done as a non-destructive method to determine the yield strength of a material, however it will not be as accurate as a proper tensile test.

Hardness is important in systems where objects rub against each other, as harder objects are more resistant to wear.

To quantify hardness, there are different tests using different indenters, and the relationship/conversion between two hardness scales is determined experimentally.



Different hardness tests are more useful for stronger or weaker materials (recall from EMATL 1)